Data management System Design

Assignment - 4

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Exercise - 1 (Constraints in SQL)

1. No faculty member from department number 5 can teach more than four courses.

A.

```
CREATE ASSERTION NoMoreThanFourCourses
CHECK (NOT EXISTS (SELECT FID, COUNT(NAME)
                   FROM FACULTY F, CLASS C
                   WHERE F.FID = C.FID AND
                         DEPTID=5
                   GROUP BY FID
                   HAVING COUNT(NAME)>4));
CHECK ((4>=ALL (SELECT COUNT(NAME)
                FROM FACULTY F, CLASS C
                WHERE F.FID = C.FID AND
                      DEPTID=5
                GROUP BY FID)));
```

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2. The number of CS majors must be more than the number of math majors.

A. CREATE ASSERTION CSmoreThanMath

```
CHECK ((SELECT COUNT(*)
        FROM STUDENTS
        WHERE S.MAJOR = 'CS')
        >
        (SELECT COUNT(*)
        FROM STUDENTS
        WHERE S.MAJOR = 'math')
        );
```

3. No Student should enroll in more than 2 classes offered by the same faculty

A. CREATE ASSERTION NomoreThan2SameFaculty

```
CHECK (NOT EXISTS (SELECT E.SUM, C.FI
        COUNT (E.CNAME)
        FROM ENROLLED E, CLASS C
        WHERE C.NAME = E.CNAME
        GROUP BY E.SNUM, C.FID
```

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```
HAVING COUNT (E.CNAME) > 2));

CHECK (2 >= ALL (SELECT COUNT (CNAME)
                FROM ENROLLED, CLASS
                WHERE NAME = CNAME
                GROUP BY SNUM, FID)));
```

## Exercise - 2 (Triggers) :-

a. Row Level Trigger

```
CREATE TRIGGER No_of_REDS
AFTER INSERT ON RESERVES
FOR EACH ROW
WHEN ((SELECT COLOR
       FROM BOATS
       WHERE BID = NEW.BID) = 'red')
UPDATE SAILORS
SET NO_OF_RED = NO_OF_RED+1
WHERE SID = NEW.SID.
```

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b. Statement level trigger

```
CREATE TRIGGER NO_OF_REDS_S
AFTER INSERT ON RESERVES
FOR EACH STATEMENT
REFERENCING NEW TABLE AS N
WHEN (EXISTS (SELECT *
              FROM BOATS, N
              WHERE N.BID =
        BOATS.BID AND COLOUR = 'red'))
UPDATE SAILORS S
SET NO_OF_RED = NO_OF_RED + (SELECT
                COUNT(*)
                FROM BOATS B, N
                WHERE N.BID = B.BID AND
                B.COLOR = 'red' AND S.SID
                = N.SID)
WHERE S.SID IN (SELECT DISTINCT N.SID
                FROM BOATS B, N
                WHERE N.BID = B.BID AND
                B.COLOR = 'red')
```

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